

TEE-RPCH Five-Experiment Preview (MNT224)

Real SGX measurements from five_experiments_20260429/data_mnt224

1 Scope

This preview PDF is generated only from the completed MNT224 real-SGX results under five_experiments_20260429. It is intended to let us inspect the corrected five-experiment structure before the A1 run finishes.

2 Experiment 1: Theory Comparison

Table 1: Theory comparison.

scheme	keygen	hash	adapt
FB-PCH	$O(S)$	$O(policy)$	$O(policy)$
EHR-RPCH	$O(S + \log N)$	$O(policy)$	$O(policy + \text{outsourced revoke})$
PNITCH	$O(S)$	$O(1)$	$O(t)$ partial adapt + combine
RPCH-XNM'21	$O(S + \log N)$	$O(policy)$	$O(policy + \log N)$
TAR-PCH	$O(S + \log N)$	$O(policy)$	$O(policy + \text{cloud update})$
TPCH	$O(S)$	$O(policy)$	$O(policy + \text{update rounds})$
TEE-RPCH (ours)	$O(S)$	$O(policy)$	Server $O(policy)$, TEE $O(\log N)$, User $O(1)$

3 Experiment 2: Baseline Comparison

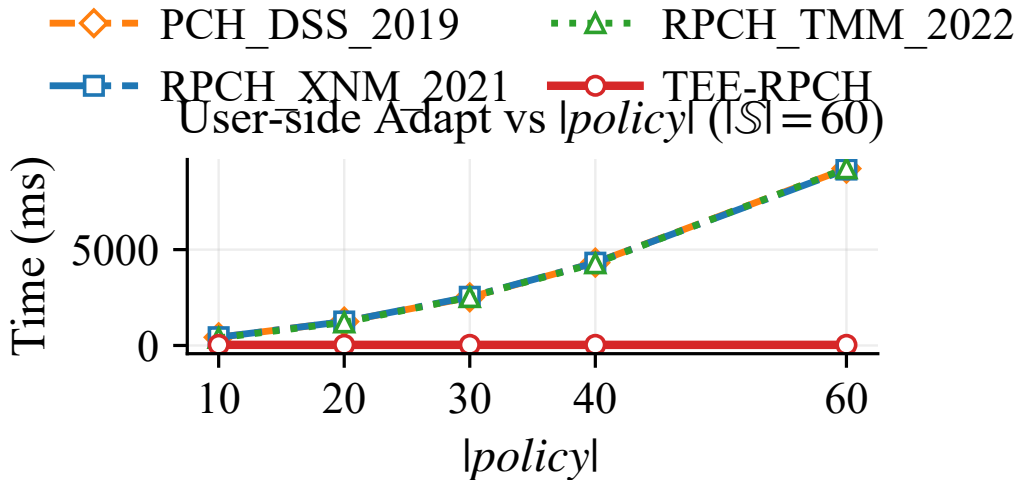


Figure 1: Online adaptation cost versus policy size.

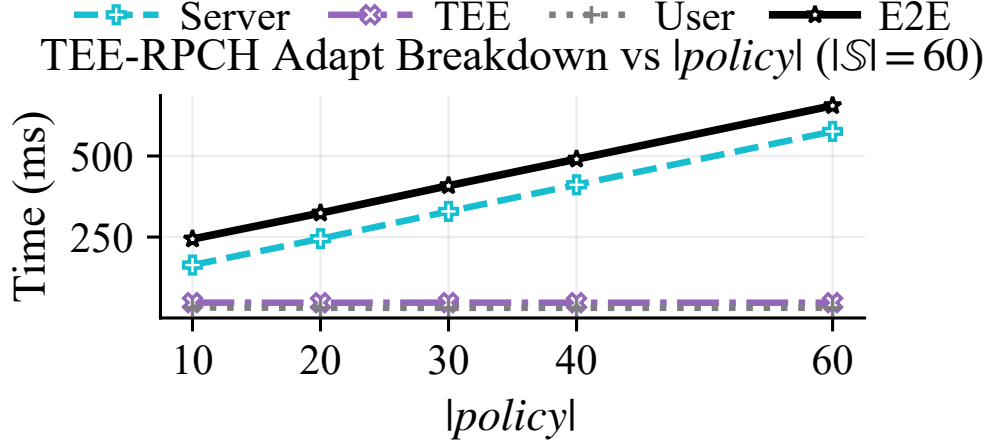


Figure 2: Server, enclave, and user breakdown.

Table 2: MNT224 baseline comparison (ms).

policy	single_outsource_ms	tee_rpch_e2e_ms	server_ms	tee_ms	user_ms
10	98.977	243.907	163.597	47.774	32.536
20	139.716	324.002	244.844	47.390	31.768
30	180.884	408.361	329.024	47.719	31.618
40	225.630	490.304	411.217	47.496	31.591
60	305.502	655.117	575.734	47.510	31.873

4 Experiment 3: Hash-Check Benefit

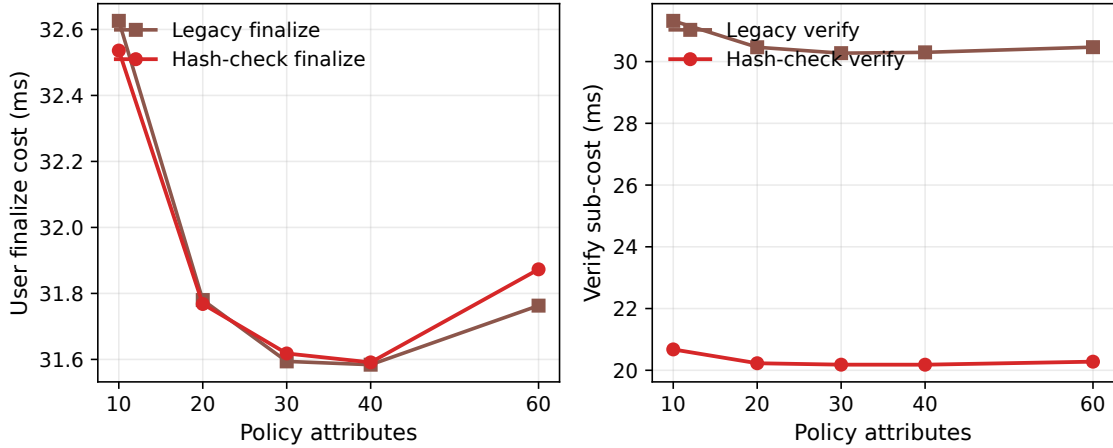


Figure 3: Legacy finalize path versus hash-check path on MNT224.

Table 3: MNT224 user-side finalize comparison (ms).

policy	legacy_user_ms	hashcheck_user_ms	legacy_verify_ms	hashcheck_verify_ms
10	32.626	32.536	31.319	20.676
20	31.780	31.768	30.464	20.228
30	31.594	31.618	30.273	20.182
40	31.584	31.591	30.299	20.181
60	31.763	31.873	30.467	20.279

5 Experiment 4: Multi-thread Evaluation

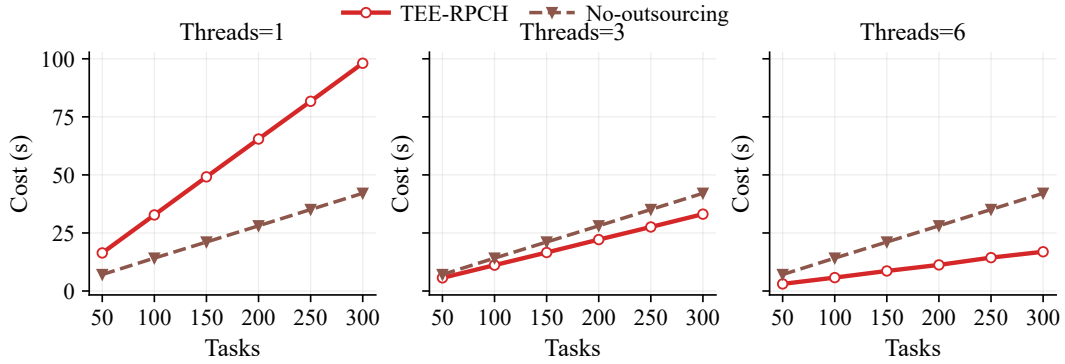


Figure 4: Wall-clock cost versus task count.

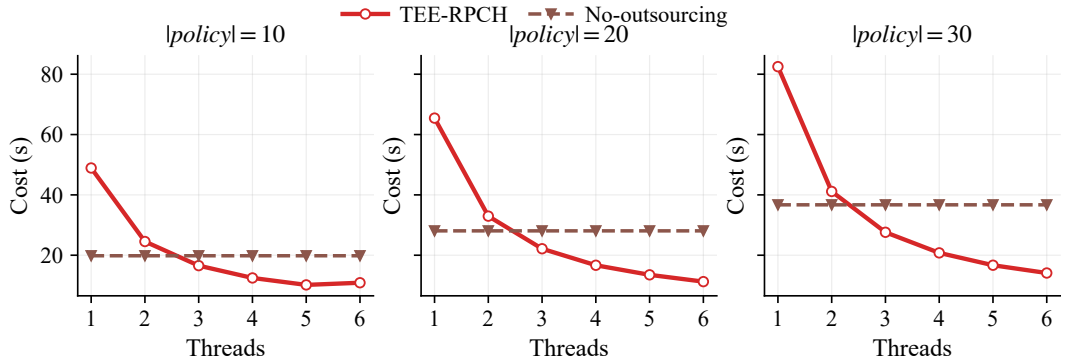


Figure 5: Wall-clock cost versus server thread count.

6 Experiment 5: Revocation

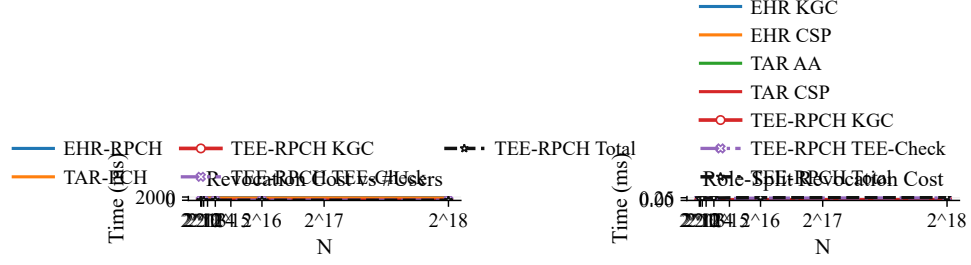


Figure 6: Revocation cost versus number of users.

Table 4: MNT224 revocation comparison (ms).

users	ehr_kgc_ms	ehr_csp_ms	ehr_total_ms	tar_aa_ms	tar_csp_ms	tar_total_ms	tee_kgc_ms	tee_ms	tee_total_ms
2 ¹⁰	0.000	2160.780	2160.780	0.001	1421.470	1421.470	0.042	0.150	0.192
2 ¹¹	0.002	2184.030	2184.030	0.001	1411.680	1411.680	0.042	0.146	0.188
2 ¹²	0.001	2167.260	2167.270	0.001	1411.420	1411.430	0.045	0.160	0.205
2 ¹³	0.000	2174.770	2174.770	0.000	1402.190	1402.190	0.047	0.175	0.222
2 ¹⁴	0.001	2177.090	2177.090	0.000	1431.640	1431.640	0.050	0.175	0.225
2 ¹⁵	0.001	2170.240	2170.250	0.000	1410.300	1410.300	0.051	0.189	0.240
2 ¹⁶	0.001	2181.990	2181.990	0.000	1406.070	1406.070	0.059	0.211	0.270
2 ¹⁷	0.001	2160.750	2160.750	0.001	1451.990	1451.990	0.060	0.224	0.284
2 ¹⁸	0.000	2167.150	2167.150	0.000	1414.210	1414.210	0.066	0.232	0.298